

D852

Astronomy

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

An observatory with a 4-meter telescope, built on a dry mountaintop, records the spectrum of a distant star. The spectrum shows dark lines at exactly the wavelengths where a lab sample of sodium gives bright lines. The star looks orange-red to the naked eye. What does this show?

- A. Sodium is present in the star's outer layers, absorbing those wavelengths.
- B. The star contains no sodium, since the lines are dark instead of bright.
- C. The star is moving toward Earth.
- D. Its orange-red color means the star is made mostly of sodium.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A hydrogen line measured at 656.3 nm in the lab appears at 656.9 nm in a star's spectrum. The star is a bright naked-eye object in a well-known constellation, and its surface temperature is about 9,000 K. What does the shift show about the star's motion?

- A. The star is moving away from Earth.
- B. The star is cooling down.
- C. The star is moving sideways across the sky.
- D. The star is moving toward Earth.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A spectral line with a lab wavelength of 500.0 nm is observed at 499.0 nm. Using $v = c \cdot \Delta\lambda / \lambda$ with $c = 300,000$ km/s, what is the star's radial motion?

- A. 600 km/s away from Earth.
- B. 0.002 km/s toward Earth.
- C. 600 km/s toward Earth.
- D. It is 600 light-years away.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Using observations taken six months apart from a space telescope, astronomers measure a parallax angle of 0.25 arcseconds for a star. The star is about twice the Sun's mass. How far away is it?

- A. 4 parsecs.
- B. 2 parsecs.
- C. 8 parsecs.
- D. 0.25 parsecs.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A star is 20 parsecs away. What parallax angle would astronomers measure?

- A. 0.5 arcseconds.
- B. 0.05 arcseconds.
- C. 0.05 light-years.
- D. 20 arcseconds.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Two stars formed in the same nebula at the same time. Star A has 10 times the Sun's mass; Star B has 0.5 times the Sun's mass. Star A is catalogued under a Greek letter and is easy to find with binoculars. Which statement about their lives is correct?

- A. Both will end as white dwarfs because they formed together.
- B. All stars end as supernovae, so both will explode.
- C. Star A will leave the main sequence far sooner and likely end in a supernova; Star B will outlive the Sun and end as a white dwarf.
- D. Star A will live longer because it has more fuel.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A main-belt asteroid discovered by an amateur team has an average distance from the Sun of 4 AU. It is about 30 km across and has a dark, carbon-rich surface. What is its orbital period?

- A. 16 years.
- B. 8 years.
- C. 64 years.
- D. 4 years.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A comet has an orbital period of 27 years. What is its average distance from the Sun?

- A. 27 AU.
- B. 9 AU.
- C. 3 AU.
- D. 5.2 AU.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A comet on a highly elliptical orbit, first recorded in old monastery records, passes perihelion at 0.6 AU and aphelion at 35 AU. Its tail is brightest near the Sun. Where in its orbit is the comet moving fastest?

- A. It moves at the same speed everywhere because its period is fixed.
- B. At aphelion, its farthest point.
- C. Midway between perihelion and aphelion.
- D. At perihelion, its closest point to the Sun.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A paleoclimate lab drilling ice cores in Antarctica, funded by an international consortium, compares past ice ages with orbital cycles. During one interval, Earth's axial tilt decreases from 24.5° to 22.1°. How does this change favor ice-sheet growth?

- A. Lower tilt moves Earth farther from the Sun.
- B. Lower tilt makes winters much colder, which is what builds ice sheets.
- C. Lower tilt gives cooler high-latitude summers, so less winter snow melts and ice accumulates.
- D. Tilt never affects climate; only orbit shape does.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

If Earth's axial tilt increased, what would happen to the difference between summer and winter at mid-latitudes?

- A. The seasonal contrast would decrease.
- B. Summers would get warmer, and winters would stay the same.
- C. No change, since the Earth-Sun distance stays the same.
- D. The seasonal contrast would increase: warmer summers and colder winters.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A museum's geology hall displays a 4.0-billion-year-old gneiss from northern Canada beside a chondrite meteorite dated at 4.56 billion years, which was found by a farmer in 1938. Why do scientists use the meteorite, not the gneiss, to estimate Earth's age?

- A. Meteorites came from another star system that is older.
- B. Earth is 4.0 billion years old, so the meteorite is irrelevant.
- C. Earth rocks cannot be radiometrically dated at all.
- D. Earth's earliest crust has been recycled by plate tectonics and erosion, while meteorites formed with the solar system and preserve its original age.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

An observatory ~~with a 4-meter telescope, built on a dry mountaintop,~~ records the spectrum of a distant star. The spectrum shows dark lines at exactly the wavelengths where a lab sample of sodium gives bright lines. ~~The star looks orange-red to the naked eye.~~ What does this show?

Target word: what the dark lines show **Decides:** Dark lines match sodium's lab wavelengths.

A	Correct	Cooler gas absorbs the same wavelengths its atoms would emit, so matching dark lines identify the element.
B	Inverted operation	Dark lines are absorption by the element, not its absence.
C	Confusion set	Motion shows up as a shift of lines from lab values; these lines match exactly.
D	Camouflage	Color tracks temperature, not composition.

2. Answer A Concept 2

A hydrogen line measured at 656.3 nm in the lab appears at 656.9 nm in a star's spectrum. ~~The star is a bright naked-eye object in a well-known constellation, and its surface temperature is about 9,000 K.~~ What does the shift show about the star's motion?

Target word: direction of motion **Decides:** Observed wavelength is longer than lab — redshift.

A	Correct	A longer observed wavelength is a redshift, meaning the source is receding.
B	Confusion set	Temperature changes the color peak, not the position of a specific line.
C	Adjacent category	Doppler shift measures only motion along the line of sight.
D	Inverted operation	Approach shortens wavelengths (blueshift).

3. Answer C Concept 2 · transfer

A spectral line with a lab wavelength of 500.0 nm is observed at 499.0 nm. Using $v = c \cdot \Delta\lambda / \lambda$ with $c = 300,000$ km/s, what is the star's radial motion?

Target word: speed and direction from a shift **Decides:** $\Delta\lambda = -1.0$ nm (blueshift).

A	Inverted operation	Right speed, wrong direction — a shorter wavelength is a blueshift.
B	Dropped element	That is $\Delta\lambda / \lambda$ alone; it drops the multiplication by c .
C	Correct	$1.0 \div 500.0 \times 300,000 = 600$ km/s; the shorter wavelength means approach.
D	Confusion set	Doppler shift gives speed along the line of sight, not distance.

4. Answer A Concept 3

Using observations taken six months apart ~~from a space telescope,~~ astronomers measure a parallax angle of 0.25 arcseconds for a star. ~~The star is about twice the Sun's mass.~~ How far away is it?

Target word: distance from parallax **Decides:** $d = 1 \div 0.25$.

A	Correct	$d = 1/p = 1/0.25 = 4$ pc (about 13 light-years).
B	Wrong stem number	Halves the result for the two observation dates; the formula already uses the half-angle.
C	Camouflage	Doubles the answer for the star's mass, which has nothing to do with parallax.
D	Inverted operation	Treats distance as equal to the angle instead of its reciprocal.

5. Answer B Concept 3 • transfer

A star is 20 parsecs away. What parallax angle would astronomers measure?

Target word: parallax from distance **Decides:** $p = 1 \div d$.

A	Wrong stem number	That is 1/2, not 1/20 — a dropped digit from the stem.
B	Correct	$p = 1/20 = 0.05''$.
C	Confusion set	Mixes up an angle with a distance unit.
D	Inverted operation	Treats angle as growing with distance; farther stars have smaller parallax.

6. Answer C Concept 4

Two stars formed in the same nebula at the same time. Star A has 10 times the Sun's mass; Star B has 0.5 times the Sun's mass. ~~Star A is catalogued under a Greek letter and is easy to find with binoculars.~~ Which statement about their lives is correct?

Target word: lifetime and fate by mass **Decides:** A is 10 solar masses; B is 0.5.

A	Adjacent category	Shared origin does not set fate; mass does.
B	Absolute language	Low-mass stars like B shed their outer layers and leave white dwarfs.
C	Correct	Luminosity rises steeply with mass, so massive stars burn out fast and die violently.
D	Inverted operation	Massive stars burn fuel so much faster that they live shorter lives.

7. Answer B Concept 5

A main-belt asteroid ~~discovered by an amateur team~~ has an average distance from the Sun of 4 AU. ~~It is about 30 km across and has a dark, carbon-rich surface.~~ What is its orbital period?

Target word: orbital period **Decides:** $P^2 = 4^3 = 64$.

A	Inverted operation	Squares the distance instead of raising it to the 3/2 power.
B	Correct	$P = \sqrt{64} = 8$ years.
C	Dropped element	Stops at $P^2 = 64$ and drops the square root.
D	Adjacent category	Assumes period grows in step with distance (a straight-line rule).

8. Answer B Concept 5 • transfer

A comet has an orbital period of 27 years. What is its average distance from the Sun?

Target word: distance from period **Decides:** $a^3 = 27^2 = 729$.

A	Adjacent category	Treats distance as equal to period.
B	Correct	$a = \sqrt[3]{729} = 9$ AU.
C	Dropped element	Takes the cube root of 27 but drops the squaring of P.
D	Inverted operation	Takes $\sqrt{27}$ — applies the wrong exponent.

9. Answer D Concept 6

A comet on a highly elliptical orbit, ~~first recorded in old monastery records~~, passes perihelion at 0.6 AU and aphelion at 35 AU. ~~Its tail is brightest near the Sun.~~ Where in its orbit is the comet moving fastest?

Target word: location of maximum speed **Decides:** Equal areas in equal times.

A	Confusion set	A fixed period does not mean constant speed on an ellipse.
B	Inverted operation	Reverses the law; it is slowest at aphelion.
C	Adjacent category	Speed rises steadily all the way to perihelion.
D	Correct	To sweep equal areas, the comet must cover more distance per unit time when close to the Sun.

10. Answer C Concept 7

A paleoclimate lab ~~drilling ice cores in Antarctica, funded by an international consortium~~, compares past ice ages with orbital cycles. During one interval, Earth's axial tilt decreases from 24.5° to 22.1°. How does this change favor ice-sheet growth?

Target word: mechanism linking lower tilt to ice **Decides:** Summer melt at high latitudes controls ice survival.

A	Confusion set	Tilt changes the angle of sunlight, not the Earth-Sun distance.
B	Adjacent category	Winter cold matters less than summer melt; lower tilt actually gives milder winters.
C	Correct	Ice sheets grow when summer is too cool to melt the previous winter's snow.
D	Absolute language	Tilt is one of the three main Milankovitch cycles.

11. Answer D Concept 7 • transfer

If Earth's axial tilt increased, what would happen to the difference between summer and winter at mid-latitudes?

Target word: effect of greater tilt **Decides:** More tilt = stronger seasonal contrast.

A	Inverted operation	That is the effect of lower tilt.
B	Dropped element	Drops the winter half; both seasons move apart.
C	Confusion set	Seasons come from tilt, not distance.
D	Correct	Greater tilt makes summer sunlight more direct and winter sunlight more slanted.

12. Answer D Concept 8

A ~~museum's geology hall~~ displays a 4.0-billion-year-old gneiss from northern Canada beside a chondrite meteorite dated at 4.56 billion years, ~~which was found by a farmer in 1938~~. Why do scientists use the meteorite, not the gneiss, to estimate Earth's age?

Target word: why meteorites date Earth **Decides:** Earth's earliest crust has been recycled.

A	Confusion set	Most meteorites are debris from our own solar system's formation.
B	Wrong stem number	Uses the oldest surviving rock as Earth's age.
C	Absolute language	Earth rocks are routinely dated; the gneiss has a date.
D	Correct	Meteorites and the Moon lack active tectonics, so they keep the oldest material.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Absorption lines reveal composition	Dark lines at specific wavelengths match the fingerprint of elements in the cooler gas the starlight passes through.	Thinking dark lines mean the element is missing, or that color alone gives composition.	Dark lines that line up with a lab element's lines.
2	Doppler shift gives radial motion	Lines shifted to longer wavelength (redshift) mean the source is moving away; shorter (blueshift) means approaching; $v = c \cdot \Delta\lambda / \lambda$.	Reading redshift as "the star is red" or "the star is cooling."	A known lab wavelength next to an observed wavelength.
3	Parallax distance	Distance in parsecs = $1 \div$ parallax in arcseconds; a smaller angle means a farther star.	Treating distance as proportional to the angle.	An angle given in arcseconds.
4	Stellar mass sets lifetime and fate	More massive stars burn fuel much faster and live shorter lives; low-mass stars end as white dwarfs, high-mass stars as supernovae leaving neutron stars or black holes.	More fuel means a longer life.	Two stars compared by mass.
5	Kepler's third law	For bodies orbiting the Sun, $P^2 = a^3$ with P in years and a in AU.	Assuming period grows in step with distance.	An orbit size and a period in the same question.
6	Kepler's second law	A line from the Sun to a planet sweeps equal areas in equal times, so an orbiting body moves fastest at perihelion and slowest at aphelion.	Constant orbital speed, or fastest when farthest.	An elliptical orbit with "where is it fastest?"
7	Milankovitch cycles and climate	Cycles in eccentricity (~100,000 yr), tilt (~41,000 yr), and precession (~26,000 yr) redistribute sunlight by season and latitude; cool high-latitude summers let ice sheets grow.	Thinking cold winters build ice sheets, or that the cycles work by changing Earth-Sun distance.	A change in tilt or orbit shape linked to ice ages.
8	Meteorites and lunar rocks date early Earth	The oldest meteorites (~4.56 billion years) date the solar system's formation; Earth's oldest rocks are younger because plate tectonics and erosion recycle crust.	Taking the oldest Earth rock as Earth's age.	An Earth rock age compared with a meteorite age.

Course scope (public description): Astronomy explores the fundamental principles of the universe, focusing on the lifecycle of stars, the structure of our solar system, and the dynamics of the Sun-Earth-Moon system. In this course, students learn about the analysis of starlight spectra and brightness to determine the composition, movement, and distance of stars. Kepler's laws are examined to explain the motions of orbiting celestial bodies and the factors that influence their paths. Students will also learn how the cyclical changes in Earth's orbit and axial orientation affect climate over geological time scales. Additionally, students will build on their geological knowledge by studying extraterrestrial objects, such as lunar rocks and meteorites, to gain insights into Earth's formation and history.